

Duo Xu

Tianjin University | State Key Laboratory of Precision Testing Technology and Instruments
Tel: +86 15242390985 | Email: 2024202134@tju.edu.cn | Website: terra11113.github.io

RESEARCH INTERESTS

- Vision-guided robotics; language-conditioned manipulation; visual feedback, calibration, and safe physical control; physics-guided computational imaging; automatic optical alignment; image-quality assessment; machine vision; learning-assisted modeling and optimization for real optical/robotic systems.

EDUCATION

**Tianjin University | M.Eng. in Electronic Information,
recommended admission**

Sep 2024 - Present

- Advisor: Prof. Xiaodong Zhang.
- Average score: 87.88/100; rank: 15/115.
- Research focus: freeform ultra-short-throw projection systems, image-quality-driven alignment, optical simulation, and closed-loop optimization.

**University of Science and Technology Beijing | B.Eng. in
Mechanical Engineering**

Sep 2020 - Jul 2024

- Average score: 83.62/100; rank: 17/178.

**University of Science and Technology Beijing | Second Major
Coursework in Computer Science and Technology**

Sep 2021 - Jul 2024

- Completed systematic training in programming, data structures, image processing, and computational approaches to engineering problems.

PUBLICATIONS AND INTELLECTUAL PROPERTY

- Jiang Li, Duo Xu, Changshuai Fang, Zexiao Li, and Xiaodong Zhang. "Sensitivity-informed multi-objective adaptive weighting method for automatic alignment of freeform USTP systems." Optics Express 34(11), 20358-20378, 2026. DOI: 10.1364/OE.34.020358. Jiang Li and Duo Xu contributed equally.
- Patent application under review: "An optimization and alignment method for imaging and display systems based on physical-model-driven adaptive weighting" (translated title).

TECHNICAL SKILLS

- Programming and engineering software: C++, Qt, Python, MATLAB, CMake, Git/GitLab; closed-loop software development, hardware communication, automation scripts, and technical documentation.
- Computer vision and image processing: OpenCV, Halcon; camera calibration basics, image preprocessing, structured target analysis, feature extraction, geometric-distortion evaluation, MTF-related image-quality assessment, and basic image reconstruction.
- Deep learning and AI infrastructure: familiar with CNN/Transformer-based vision models, common detection/backbone architectures, basic model training/deployment workflows, CUDA fundamentals; PyTorch and Hugging Face.
- Optical/physical modeling and simulation: Zemax OpticStudio, ZOS-API, FDTD Lumerical; non-sequential ray tracing, 6-DOF perturbation analysis, tolerance/sensitivity modeling, optical-field simulation, and simulation-to-experiment validation.
- Robotics and hardware-oriented implementation: LeRobot, SO-101 leader-follower teleoperation, real-robot data collection, ACT, SmolVLA, multimodal rollout evaluation; industrial cameras, 6-DOF motion platforms, HLS/FPGA adaptation, TCP/UDP, SolidWorks, AutoCAD, and Ansys Workbench.

RESEARCH AND PROJECT EXPERIENCE

**Automatic Alignment and Image-quality Optimization of
Freeform USTP Systems**

2024.11 - Present

Tianjin University | Master's research project

- Defined the task as a vision-feedback closed-loop alignment problem: camera-captured target images are used to infer optical pose errors and guide a 6-DOF precision platform.
- Built a Zemax non-sequential model matched to the physical prototype, scanned 6-DOF freeform-mirror perturbations, and summarized pose-to-image-quality response laws for geometric distortion and MTF.
- Designed the structured-target measurement workflow and implemented a C++ adaptive weighting optimizer; integrated simulation, industrial-camera acquisition, image-quality evaluation, and motion execution.
- Validated the full system in repeated experiments: final mean MTF exceeded 70%, each alignment run took about 10 minutes, and result dispersion was reduced.

**Real-Robot Language-Conditioned Manipulation and VLA
Failure Analysis**

2026.7 - Present

SO-101 / LeRobot research platform

- Built and calibrated a dual-view SO-101 leader-follower platform with external and wrist RGB cameras; implemented teleoperation, synchronized RGB/state/action recording, and real-robot rollout workflows.
- Defined a repeatable tabletop protocol with three color-conditioned pick-and-place instructions, fixed workspace and camera settings, and consistent task-success and failure labels.
- Curated 60 dual-view real-robot episodes and performed an episode-level 48/6/6 train/validation/test split, preserving the separation between demonstrations and held-out evaluation.
- Established an ACT visual-imitation baseline with 8/10 real-robot success on a fixed pick-and-place task, validating the sensing, data, training, and deployment loop before multi-task VLA evaluation.

- Fine-tuned and deployed SmolVLA for the three-instruction setting; documented target-directed rollout motion together with incomplete descend-grasp-lift-place failures for subsequent data and observability analysis.

Wavefront-aberration-guided Automatic Alignment of Optical Lens Groups

2025.5 - Present

Tianjin University | Collaborative research project with Huawei Terminal BG

- Extended alignment feedback from MTF/PSF toward wavefront-aberration information, aiming to improve misalignment estimation when conventional image-quality feedback is insufficient.
- Participated in mapping low-order Zernike coefficients to lens-group misalignment parameters, and developed Qt/C++ modules based on ZOS-API for Zemax automation, perturbation scanning, ray tracing, and result export.
- Supported phase-deflectometry experiments for off-axis optical components, including acquisition workflow debugging, data processing, simulation-experiment comparison, and error analysis.

Chromatic-aberration-enhanced Metalens Design and Simulation for Spectral Confocal Sensing

2024.3 - 2024.10

Tianjin University and USTB | Undergraduate thesis project

- Designed a metasurface-based solution for spectral confocal displacement sensing, exploring whether a metalens could replace part of a traditional multi-lens dispersion structure.
- Derived wavelength-dependent phase profiles and meta-atom rotation rules; used FDTD Lumerical and MATLAB to screen nanofin meta-atoms and evaluate focal and optical-field distributions.
- Connected optical-field simulation with sensing-system design rather than treating imaging components as isolated devices.

FPGA-based Line-scan Vision Inspection System

2024.9 - 2024.10

Tianjin University | Collaborative research project with Hexagon

- Developed the software-side image-processing workflow for high-speed line-scan inspection, balancing field of view, resolution, and real-time throughput.
- Built C++ modules for reconstruction, preprocessing, feature extraction, candidate-region analysis, and detection-result output; adapted selected algorithms into HLS/FPGA-friendly processing blocks.
- Participated in software-hardware co-debugging and deployment-oriented evaluation under streaming data and memory-access constraints.

RESEARCH STRENGTHS AND PREPARATION

- Vision-feedback closed-loop system building: experienced in connecting target design, camera acquisition, image preprocessing, metric extraction, optimization, and motion execution on a real platform.
- Physics-to-algorithm modeling: able to translate optical response laws, tolerance behavior, and image-quality degradation patterns into optimization objectives, decision rules, and control-relevant feedback.
- Calibration and experimental validation: prepared to extend optical alignment experience toward robotic calibration, visual servoing, state estimation, real-robot dataset discipline, and precision operation.
- Cross-disciplinary engineering preparation: strong foundation across optics, mechanical systems, image processing, C++ implementation, learning-based vision, and hardware-aware deployment.

ENGLISH PROFICIENCY

- IELTS Academic: Overall 6.5 (Listening 7.5, Reading 7.0, Writing 6.5, Speaking 5.5). CET-6 and CET-4 passed.

SELECTED AWARDS AND HONORS

- Second-Class Academic Scholarship, Tianjin University, 2025.
- People's Scholarship, University of Science and Technology Beijing, 2021, 2022, and 2023.
- Outstanding Student Cadre, University of Science and Technology Beijing, 2022 and 2023.
- Third Prize, National College Students Mathematics Competition, 2021; Third Prize, National English Competition for College Students, 2022.